

CÉSAR ALEXANDER DELGADILLO SÁNCHEZ

Software & Data Engineer

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SUMMARY

Software and Data Engineer experienced in building reliable cloud-based data workflows and operational systems across Microsoft Azure and Google Cloud Platform. Skilled in SQL Server, BigQuery, Python, REST APIs, Samsara telematics, CDC replication, batch processing, data quality, and production automation with Azure Functions and GitHub Actions.

EXPERIENCE

TransNort

Software & Data Engineer

July 2026 – Present

- Engineered operational data workflows integrating **Samsara telematics, SQL-based services, REST APIs, and Azure infrastructure** to support fleet monitoring, operational analytics, and automated reporting.
- Improved **data consistency and freshness across distributed Azure workloads** by implementing reliable cache invalidation, authoritative reads, concurrency handling, and synchronization between transactional writes and downstream read models.
- Designed robust **batch processing and data-import workflows** using idempotency, deduplication, canonical identifiers, preflight validation, and transactional safeguards to prevent duplicate or partially committed operational data.
- Built automated **data processing and reporting pipelines** that classify, aggregate, and enrich Samsara and SQL data, orchestrating scheduled production jobs with **Azure Functions and GitHub Actions** and implementing failure recovery and deduplication.

Grupo DEACERO

Monterrey, México

Developer Analyst

May 2026 – July 2026

- Diagnosed production application issues by tracing **SQL Server** execution flows, identifying data inconsistencies and blocked process states, and applying controlled SQL corrections to restore operational processes.

Data Engineer Intern

March 2025 – May 2026

- Led the migration of mission-critical **ETL workflows** from **SQL Server** to **Google BigQuery**, refactoring legacy **T-SQL** logic and building a **Python-based** QA framework to validate data consistency across large-scale inventory datasets.
- Designed and implemented a **Salesforce-to-BigQuery replication architecture** using **Google Datastream**, automated dataset validation, and dynamic object streaming through **GCP REST APIs**, reducing data latency to near real-time levels.
- Built a reusable multi-source replication module for **SQL Server** and **Salesforce**, improving code maintainability, error handling, and extensibility across replication workflows.
- Operated and monitored **CDC-based replication pipelines** for SQL-to-SQL and SQL-to-BigQuery synchronization, ensuring reliable data consistency across environments.

TECHNICAL SKILLS

- Cloud & Data:** Google Cloud Platform, Microsoft Azure, Azure Functions, BigQuery, Google Datastream, Samsara Telematics, SQL Server, ETL Pipelines, Batch Processing, CDC, Data Quality, Apache Airflow, Docker, Power BI.
- Programming Languages:** Python, SQL, C#, C, TypeScript, JavaScript, HTML/CSS.
- Backend & APIs:** RESTful Architecture, JWT Authentication, Microservices, Cache Invalidation, Concurrency Handling, Supabase implementation.
- Frameworks & Libraries:** .NET Core, Angular, Ionic, TailwindCSS, Astro, React.
- Tools:** Git/GitHub, GitHub Actions, Postman, Firebase, SQL Server, MySQL.
- AI & Automation:** AI-Pair Programming, Prompt Engineering, LLM Integration.
- Languages:** Spanish (Native), English (C1 - Professional Proficiency).

PROJECTS

Saldito | Astro, TypeScript, FastAPI, SQLAlchemy, PostgreSQL, Supabase

- Built a mobile-first personal finance application using **FastAPI**, **SQLAlchemy**, **PostgreSQL**, **Supabase**, **Astro**, and **TypeScript** to manage accounts, transactions, budgets, and savings goals.
- Designed **relational data models for accounts**, credit cards, installment purchases, recurring payments, transfers, and financial reports.
- Implemented cash-flow forecasting, CSV import/export, receipt uploads, and spending analytics to provide users with structured financial insights.

PSell Kitchen Dashboard | Supabase, PowerSync

- Designed a real-time, bidirectional sync architecture using **Supabase Realtime** and **PowerSync**, keeping order state consistent across web and mobile clients with sub-second latency.
- Built a Kanban-style kitchen operations dashboard app and webapp with automated visual alerts and time-based status tracking to improve throughput and reduce response to near realtime.

EDUCATION

Universidad Autónoma de Nuevo León (UANL) - FCFM

Monterrey, México

Bachelor of Science in Computer Science

- Certifications:** Advanced Web Development Applications (**US-IT**), Basic Data Science (**Google**).
- Activities:** Participant at Gran PremioMX (Algorithm & Problem Solving Competition).